

Solução do Problema nesm - Solver HiGHS

Análise Computacional

6 de outubro de 2025

1 Informações do Problema

Informações do Problema:

- Nome: nesm
- Número de Variáveis: 2923
- Número de Restrições: 662
- Inviabilidade Primal: 1.000e-08
- Inviabilidade Dual: 1.819e-12
- Valor Primal: 1.408e+07
- Valor Dual: 1.113e+08
- Gap: 2.444e-22
- Número de Iterações: 38

2 Variáveis Primais e Custos Reduzidos

Tabela 1: Variáveis primais (x) e custos reduzidos (z) do problema nesm

Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
1	1.304e+03	1	0.000e+00
2	1.533e+03	2	-2.174e-05
3	1.533e+03	3	-2.174e-05
4	9.618e+02	4	0.000e+00
5	1.532e+03	5	-1.605e-04
6	9.904e+02	6	0.000e+00
7	1.289e+03	7	0.000e+00
8	1.282e+03	8	0.000e+00
9	0.000e+00	9	1.836e-01
10	0.000e+00	10	1.833e-01
11	0.000e+00	11	1.832e-01
12	0.000e+00	12	1.835e-01

Continua na próxima página

Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
13	0.000e+00	13	1.690e-01
14	0.000e+00	14	1.761e-01
15	0.000e+00	15	1.877e-01
16	0.000e+00	16	1.877e-01
17	0.000e+00	17	9.513e-03
18	0.000e+00	18	8.144e-03
19	0.000e+00	19	8.144e-03
20	0.000e+00	20	8.166e-03
21	0.000e+00	21	6.457e-03
22	0.000e+00	22	8.625e-03
23	0.000e+00	23	9.513e-03
24	0.000e+00	24	8.166e-03
25	0.000e+00	25	5.107e-03
26	0.000e+00	26	5.950e-03
27	0.000e+00	27	5.950e-03
28	0.000e+00	28	5.981e-03
29	0.000e+00	29	2.353e-03
30	0.000e+00	30	7.771e-04
31	0.000e+00	31	8.525e-04
32	0.000e+00	32	1.669e-03
33	2.352e+02	33	-5.307e-03
34	2.751e+02	34	-6.243e-03
35	2.751e+02	35	-6.243e-03
36	2.751e+02	36	-5.924e-03
37	4.515e+02	37	-2.710e-02
38	6.751e+02	38	-2.771e-03
39	1.880e+02	39	0.000e+00
40	1.788e+02	40	0.000e+00
41	1.179e+03	41	0.000e+00
42	1.380e+03	42	0.000e+00
43	1.694e+03	43	0.000e+00
44	1.066e+03	44	0.000e+00
45	1.663e+03	45	0.000e+00
46	2.018e+03	46	0.000e+00
47	1.414e+03	47	0.000e+00
48	1.601e+03	48	0.000e+00
49	1.218e+02	49	-6.061e-03
50	1.228e+02	50	-6.607e-03
51	1.228e+02	51	-6.607e-03
52	1.228e+02	52	-6.131e-03
53	1.197e+02	53	-3.541e-02
54	9.660e+01	54	-3.294e-02
55	1.218e+02	55	-1.502e-02
56	1.228e+02	56	-2.558e-03
57	1.820e+01	57	0.000e+00
58	5.880e+01	58	-5.669e-04
59	5.880e+01	59	-5.669e-04
60	2.697e-01	60	0.000e+00
61	4.830e+01	61	-3.145e-02

Continua na próxima página

Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
62	6.559e+00	62	0.000e+00
63	2.338e+01	63	0.000e+00
64	2.163e+01	64	0.000e+00
65	0.000e+00	65	4.580e-01
66	0.000e+00	66	1.347e-01
67	0.000e+00	67	3.175e-01
68	0.000e+00	68	4.762e-01
69	0.000e+00	69	5.721e-02
70	0.000e+00	70	4.642e-02
71	0.000e+00	71	4.642e-02
72	0.000e+00	72	4.663e-02
73	0.000e+00	73	7.198e-03
74	1.775e+01	74	0.000e+00
75	0.000e+00	75	6.349e-02
76	0.000e+00	76	5.291e-02
77	0.000e+00	77	4.255e-03
78	1.777e+01	78	0.000e+00
79	5.565e+01	79	-5.990e-04
80	5.245e+01	80	0.000e+00
81	5.250e+01	81	-3.606e-02
82	6.876e+00	82	0.000e+00
83	0.000e+00	83	1.486e-02
84	0.000e+00	84	1.111e-02
85	8.043e+02	85	-1.352e+03
86	8.064e+02	86	-1.397e+03
87	8.064e+02	87	-1.397e+03
88	8.064e+02	88	-1.352e+03
89	7.528e+02	89	-1.404e+03
90	8.085e+02	90	-1.389e+03
91	8.043e+02	91	-1.382e+03
92	8.064e+02	92	-1.382e+03
93	6.825e+01	93	-5.890e-03
94	6.825e+01	94	-6.575e-03
95	6.825e+01	95	-6.575e-03
96	6.825e+01	96	-5.890e-03
97	6.825e+01	97	-2.268e-02
98	6.825e+01	98	-2.227e-02
99	0.000e+00	99	6.018e-03
100	0.000e+00	100	5.923e-03
101	6.300e+00	101	-1.351e+03
102	6.300e+00	102	-1.397e+03
103	6.300e+00	103	-1.397e+03
104	6.300e+00	104	-1.351e+03
105	7.350e+00	105	-1.404e+03
106	7.350e+00	106	-1.389e+03
107	6.300e+00	107	-1.382e+03
108	6.300e+00	108	-1.382e+03
109	0.000e+00	109	1.644e-01
110	0.000e+00	110	2.440e-01

Continua na próxima página

Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
111	0.000e+00	111	2.431e-01
112	0.000e+00	112	2.429e-01
113	0.000e+00	113	3.146e-01
114	0.000e+00	114	3.123e-01
115	0.000e+00	115	1.536e-01
116	0.000e+00	116	2.329e-01
117	0.000e+00	117	6.981e-02
118	0.000e+00	118	4.614e-02
119	0.000e+00	119	4.614e-02
120	0.000e+00	120	4.646e-02
121	0.000e+00	121	2.088e-01
122	0.000e+00	122	1.318e-01
123	0.000e+00	123	7.511e-02
124	0.000e+00	124	5.239e-02
125	0.000e+00	125	2.915e-02
126	0.000e+00	126	1.714e-02
127	0.000e+00	127	1.714e-02
128	0.000e+00	128	1.761e-02
129	3.360e+01	129	-1.359e-02
130	0.000e+00	130	4.735e-03
131	0.000e+00	131	1.189e-02
132	1.759e+00	132	0.000e+00
133	3.887e-01	133	0.000e+00
134	1.365e+01	134	-8.751e-03
135	1.365e+01	135	-8.751e-03
136	1.365e+01	136	-6.105e-03
137	0.000e+00	137	1.939e-02
138	0.000e+00	138	7.576e-02
139	0.000e+00	139	3.608e-02
140	0.000e+00	140	2.949e-02
141	4.515e+01	141	-1.450e-02
142	3.990e+01	142	-1.281e-02
143	3.990e+01	143	-1.281e-02
144	3.990e+01	144	-1.158e-02
145	7.140e+01	145	-9.838e-02
146	9.240e+01	146	-1.003e-01
147	0.000e+00	147	2.215e-02
148	0.000e+00	148	2.506e-02
149	0.000e+00	149	8.173e-01
150	0.000e+00	150	8.128e-01
151	0.000e+00	151	8.180e-01
152	0.000e+00	152	8.223e-01
153	0.000e+00	153	4.130e-01
154	0.000e+00	154	1.100e-01
155	0.000e+00	155	9.037e-01
156	0.000e+00	156	9.031e-01
157	4.399e+02	157	-1.352e+03
158	5.491e+02	158	-1.397e+03
159	5.491e+02	159	-1.397e+03

Continua na próxima página

Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
160	5.491e+02	160	-1.352e+03
161	6.762e+02	161	-1.404e+03
162	2.866e+02	162	-1.389e+03
163	4.399e+02	163	-1.382e+03
164	5.491e+02	164	-1.382e+03
165	5.040e+01	165	-1.352e+03
166	6.300e+01	166	-1.397e+03
167	6.300e+01	167	-1.397e+03
168	6.300e+01	168	-1.352e+03
169	6.405e+01	169	-1.404e+03
170	5.355e+01	170	-1.389e+03
171	5.040e+01	171	-1.382e+03
172	6.300e+01	172	-1.382e+03
173	0.000e+00	173	6.711e-02
174	0.000e+00	174	3.189e-02
175	0.000e+00	175	3.189e-02
176	0.000e+00	176	3.192e-02
177	0.000e+00	177	5.009e-02
178	0.000e+00	178	4.773e-02
179	0.000e+00	179	6.285e-02
180	0.000e+00	180	2.761e-02
181	0.000e+00	181	1.618e-02
182	0.000e+00	182	2.992e-03
183	0.000e+00	183	2.992e-03
184	0.000e+00	184	3.311e-03
185	5.355e+01	185	-1.064e-02
186	0.000e+00	186	2.749e-02
187	0.000e+00	187	2.149e-02
188	0.000e+00	188	9.235e-03
189	0.000e+00	189	2.014e-02
190	0.000e+00	190	2.013e-02
191	0.000e+00	191	2.013e-02
192	0.000e+00	192	2.014e-02
193	0.000e+00	193	2.347e-03
194	0.000e+00	194	1.472e-03
195	0.000e+00	195	5.313e-03
196	0.000e+00	196	1.727e-02
197	0.000e+00	197	8.957e-03
198	0.000e+00	198	3.538e-02
199	0.000e+00	199	3.538e-02
200	0.000e+00	200	3.585e-02
201	2.310e+01	201	-4.757e-04
202	6.960e-01	202	0.000e+00
203	3.608e+01	203	0.000e+00
204	0.000e+00	204	3.943e-02
205	0.000e+00	205	2.944e-02
206	0.000e+00	206	1.649e-02
207	0.000e+00	207	1.649e-02
208	0.000e+00	208	1.710e-02

Continua na próxima página

Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
209	1.995e+01	209	-5.610e-03
210	0.000e+00	210	4.329e-02
211	0.000e+00	211	4.762e-02
212	0.000e+00	212	3.527e-02
213	4.410e+01	213	-2.084e-02
214	6.405e+01	214	-2.935e-02
215	6.405e+01	215	-2.935e-02
216	6.405e+01	216	-2.790e-02
217	4.515e+01	217	-9.024e-02
218	5.880e+01	218	-9.416e-02
219	2.281e+01	219	0.000e+00
220	1.640e+01	220	0.000e+00
221	0.000e+00	221	1.279e-01
222	0.000e+00	222	9.741e-02
223	0.000e+00	223	9.741e-02
224	0.000e+00	224	9.768e-02
225	0.000e+00	225	1.142e-01
226	0.000e+00	226	8.658e-02
227	0.000e+00	227	1.361e-01
228	0.000e+00	228	1.058e-01
229	5.775e+01	229	-1.352e+03
230	5.565e+01	230	-1.397e+03
231	5.565e+01	231	-1.397e+03
232	5.565e+01	232	-1.352e+03
233	7.140e+01	233	-1.404e+03
234	6.090e+01	234	-1.389e+03
235	5.775e+01	235	-1.382e+03
236	5.565e+01	236	-1.382e+03
237	0.000e+00	237	1.120e-02
238	0.000e+00	238	1.052e-02
239	1.086e+01	239	0.000e+00
240	0.000e+00	240	1.024e-03
241	2.084e+01	241	0.000e+00
242	0.000e+00	242	2.000e-08
243	6.518e-01	243	0.000e+00
244	6.730e-01	244	0.000e+00
245	0.000e+00	245	1.645e+00
246	0.000e+00	246	1.526e+00
247	0.000e+00	247	3.176e-01
248	0.000e+00	248	3.176e-01
249	0.000e+00	249	3.180e-01
250	0.000e+00	250	3.175e-01
251	0.000e+00	251	4.762e-01
252	0.000e+00	252	3.175e-01
253	0.000e+00	253	4.053e-01
254	0.000e+00	254	9.481e-01
255	0.000e+00	255	1.360e-01
256	1.365e+01	256	-2.452e+00
257	1.365e+01	257	-2.537e+00

Continua na próxima página

Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
258	1.365e+01	258	-2.570e-02
259	1.365e+01	259	-2.250e-02
260	2.625e+01	260	-6.158e-02
261	3.570e+01	261	-7.058e-02
262	1.365e+01	262	-2.533e-02
263	1.365e+01	263	-2.533e-02
264	2.100e+00	264	-4.772e-01
265	2.100e+00	265	-4.927e-01
266	7.213e+02	266	0.000e+00
267	1.052e+03	267	-3.168e-05
268	1.052e+03	268	-3.168e-05
269	7.135e+02	269	0.000e+00
270	9.345e+02	270	-9.559e-05
271	9.660e+02	271	-1.178e-04
272	6.082e+02	272	0.000e+00
273	8.749e+02	273	0.000e+00
274	0.000e+00	274	1.002e-01
275	0.000e+00	275	1.003e-01
276	0.000e+00	276	1.003e-01
277	0.000e+00	277	1.002e-01
278	0.000e+00	278	9.769e-02
279	0.000e+00	279	9.508e-02
280	0.000e+00	280	9.508e-02
281	0.000e+00	281	9.507e-02
282	0.000e+00	282	1.306e-02
283	0.000e+00	283	1.633e-02
284	0.000e+00	284	1.633e-02
285	0.000e+00	285	1.637e-02
286	0.000e+00	286	1.910e-02
287	0.000e+00	287	1.005e-02
288	0.000e+00	288	1.306e-02
289	0.000e+00	289	1.637e-02
290	0.000e+00	290	3.165e-01
291	0.000e+00	291	1.049e-01
292	0.000e+00	292	1.049e-01
293	0.000e+00	293	1.049e-01
294	0.000e+00	294	1.030e-01
295	0.000e+00	295	1.309e-01
296	0.000e+00	296	3.123e-01
297	0.000e+00	297	1.006e-01
298	0.000e+00	298	9.594e-02
299	0.000e+00	299	3.331e-03
300	0.000e+00	300	3.331e-03
301	0.000e+00	301	4.101e-03
302	2.310e+01	302	-2.758e-02
303	0.000e+00	303	1.016e-01
304	0.000e+00	304	1.148e-01
305	0.000e+00	305	2.291e-02
306	0.000e+00	306	1.606e-02

Continua na próxima página

Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
307	0.000e+00	307	1.301e-03
308	0.000e+00	308	1.301e-03
309	0.000e+00	309	1.521e-03
310	5.565e+01	310	-2.328e-03
311	1.134e+02	311	-1.126e-02
312	0.000e+00	312	2.599e-03
313	7.192e+01	313	0.000e+00
314	9.450e+01	314	-4.283e-03
315	6.510e+01	315	0.000e+00
316	4.785e+01	316	0.000e+00
317	0.000e+00	317	4.955e-04
318	1.218e+02	318	-2.488e-03
319	1.134e+02	319	-1.764e-03
320	7.910e+01	320	0.000e+00
321	0.000e+00	321	4.663e-03
322	0.000e+00	322	8.488e-04
323	0.000e+00	323	5.045e-02
324	0.000e+00	324	5.045e-02
325	0.000e+00	325	5.102e-02
326	3.349e+01	326	0.000e+00
327	0.000e+00	327	2.751e-02
328	0.000e+00	328	1.669e-02
329	0.000e+00	329	5.572e-02
330	1.217e+01	330	0.000e+00
331	1.575e+01	331	-6.803e-03
332	1.575e+01	332	-6.803e-03
333	1.575e+01	333	-4.535e-03
334	4.830e+01	334	-4.279e-02
335	4.200e+01	335	-3.899e-02
336	0.000e+00	336	5.225e-03
337	1.557e+01	337	0.000e+00
338	0.000e+00	338	4.516e-02
339	0.000e+00	339	9.264e-01
340	0.000e+00	340	1.794e-02
341	0.000e+00	341	4.200e-02
342	0.000e+00	342	4.055e-02
343	0.000e+00	343	4.006e-02
344	0.000e+00	344	4.455e-02
345	0.000e+00	345	4.181e-02
346	0.000e+00	346	4.056e-02
347	0.000e+00	347	3.806e-02
348	0.000e+00	348	3.673e-02
349	0.000e+00	349	9.638e-02
350	0.000e+00	350	1.119e-01
351	0.000e+00	351	1.140e-01
352	0.000e+00	352	9.554e-02
353	0.000e+00	353	2.732e-02
354	0.000e+00	354	4.692e-02
355	0.000e+00	355	9.143e-02

Continua na próxima página

Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
356	0.000e+00	356	1.225e-01
357	0.000e+00	357	6.746e-02
358	0.000e+00	358	6.781e-02
359	0.000e+00	359	6.650e-02
360	0.000e+00	360	6.512e-02
361	0.000e+00	361	5.552e-02
362	0.000e+00	362	4.913e-02
363	0.000e+00	363	6.000e-02
364	0.000e+00	364	5.820e-02
365	0.000e+00	365	2.761e-01
366	0.000e+00	366	2.633e-01
367	0.000e+00	367	2.793e-01
368	0.000e+00	368	2.932e-01
369	0.000e+00	369	2.260e-01
370	0.000e+00	370	2.604e-01
371	0.000e+00	371	2.767e-01
372	0.000e+00	372	2.898e-01
373	0.000e+00	373	1.372e+00
374	0.000e+00	374	1.480e+00
375	0.000e+00	375	1.368e+00
376	0.000e+00	376	1.283e+00
377	0.000e+00	377	1.310e+00
378	0.000e+00	378	1.873e+00
379	0.000e+00	379	1.557e+00
380	0.000e+00	380	1.470e+00
381	0.000e+00	381	3.849e+00
382	0.000e+00	382	3.988e+00
383	0.000e+00	383	3.845e+00
384	0.000e+00	384	3.627e+00
385	0.000e+00	385	4.062e+00
386	0.000e+00	386	4.523e+00
387	0.000e+00	387	4.975e+00
388	0.000e+00	388	4.667e+00
389	0.000e+00	389	1.950e+00
390	0.000e+00	390	2.514e+00
391	0.000e+00	391	1.610e+00
392	0.000e+00	392	1.904e+00
393	0.000e+00	393	1.060e+00
394	0.000e+00	394	8.433e-01
395	0.000e+00	395	1.788e+00
396	0.000e+00	396	2.123e+00
397	0.000e+00	397	1.835e-01
398	0.000e+00	398	1.804e-01
399	0.000e+00	399	1.551e-01
400	0.000e+00	400	1.583e-01
401	0.000e+00	401	1.188e-01
402	0.000e+00	402	1.836e-01
403	0.000e+00	403	2.100e-01
404	0.000e+00	404	1.870e-01

Continua na próxima página

Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
405	0.000e+00	405	4.690e-02
406	0.000e+00	406	4.662e-02
407	0.000e+00	407	4.688e-02
408	0.000e+00	408	4.717e-02
409	0.000e+00	409	4.004e-02
410	0.000e+00	410	4.012e-02
411	0.000e+00	411	3.808e-02
412	0.000e+00	412	3.584e-02
413	0.000e+00	413	1.258e-01
414	0.000e+00	414	1.343e-01
415	0.000e+00	415	1.353e-01
416	0.000e+00	416	1.191e-01
417	0.000e+00	417	6.690e-02
418	0.000e+00	418	9.334e-02
419	0.000e+00	419	1.304e-01
420	0.000e+00	420	1.469e-01
421	0.000e+00	421	7.246e-01
422	0.000e+00	422	7.102e-01
423	0.000e+00	423	7.092e-01
424	0.000e+00	424	6.529e-01
425	0.000e+00	425	4.121e-01
426	0.000e+00	426	3.181e-01
427	0.000e+00	427	5.705e-01
428	0.000e+00	428	5.859e-01
429	0.000e+00	429	5.079e-01
430	0.000e+00	430	5.182e-01
431	0.000e+00	431	4.598e-01
432	0.000e+00	432	4.967e-01
433	0.000e+00	433	4.697e-01
434	0.000e+00	434	5.798e-01
435	0.000e+00	435	5.504e-01
436	0.000e+00	436	5.436e-01
437	0.000e+00	437	2.119e+00
438	0.000e+00	438	2.486e+00
439	0.000e+00	439	1.944e+00
440	0.000e+00	440	1.992e+00
441	0.000e+00	441	2.079e+00
442	0.000e+00	442	2.851e+00
443	0.000e+00	443	2.200e+00
444	0.000e+00	444	2.257e+00
445	0.000e+00	445	2.321e+00
446	0.000e+00	446	2.340e+00
447	0.000e+00	447	2.565e+00
448	0.000e+00	448	2.466e+00
449	0.000e+00	449	2.355e+00
450	0.000e+00	450	3.232e+00
451	0.000e+00	451	3.068e+00
452	0.000e+00	452	3.009e+00
453	0.000e+00	453	9.090e-01

Continua na próxima página

Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
454	0.000e+00	454	9.144e-01
455	0.000e+00	455	6.470e-01
456	0.000e+00	456	4.670e-01
457	0.000e+00	457	1.904e-01
458	0.000e+00	458	1.913e-01
459	0.000e+00	459	7.562e-01
460	0.000e+00	460	7.338e-01
461	0.000e+00	461	4.120e-03
462	1.312e+01	462	0.000e+00
463	3.825e+01	463	0.000e+00
464	0.000e+00	464	1.868e-03
465	0.000e+00	465	1.033e-01
466	0.000e+00	466	1.406e-01
467	0.000e+00	467	1.238e-01
468	0.000e+00	468	9.466e-02
469	0.000e+00	469	9.749e-02
470	0.000e+00	470	1.092e-01
471	0.000e+00	471	1.120e-01
472	0.000e+00	472	1.064e-01
473	0.000e+00	473	1.078e-01
474	0.000e+00	474	1.127e-01
475	0.000e+00	475	9.622e-02
476	0.000e+00	476	1.033e-01
477	0.000e+00	477	3.555e-01
478	0.000e+00	478	3.125e-01
479	0.000e+00	479	2.700e-01
480	0.000e+00	480	3.587e-01
481	0.000e+00	481	3.769e-01
482	0.000e+00	482	3.432e-01
483	0.000e+00	483	3.742e-01
484	0.000e+00	484	3.401e-01
485	0.000e+00	485	8.208e-01
486	0.000e+00	486	1.057e+00
487	0.000e+00	487	1.027e+00
488	0.000e+00	488	1.544e+00
489	0.000e+00	489	1.164e+00
490	0.000e+00	490	8.038e-01
491	0.000e+00	491	8.400e-01
492	0.000e+00	492	1.018e+00
493	0.000e+00	493	1.240e+00
494	0.000e+00	494	2.815e+00
495	0.000e+00	495	1.538e+00
496	0.000e+00	496	1.480e+00
497	0.000e+00	497	2.821e+00
498	0.000e+00	498	2.293e+00
499	0.000e+00	499	1.494e+00
500	0.000e+00	500	1.845e+00
501	0.000e+00	501	2.683e+00
502	0.000e+00	502	3.035e+00

Continua na próxima página

Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
503	0.000e+00	503	1.933e+00
504	0.000e+00	504	2.524e+00
505	0.000e+00	505	1.900e+00
506	0.000e+00	506	2.864e+00
507	0.000e+00	507	2.289e+00
508	0.000e+00	508	2.595e+00
509	0.000e+00	509	3.656e-01
510	0.000e+00	510	3.702e-01
511	0.000e+00	511	3.747e-01
512	0.000e+00	512	3.888e-01
513	0.000e+00	513	3.192e-01
514	0.000e+00	514	4.690e-01
515	0.000e+00	515	4.770e-01
516	0.000e+00	516	4.822e-01
517	0.000e+00	517	1.036e-01
518	0.000e+00	518	1.040e-01
519	0.000e+00	519	9.614e-02
520	0.000e+00	520	9.806e-02
521	3.126e+02	521	0.000e+00
522	0.000e+00	522	7.288e-03
523	0.000e+00	523	1.248e-01
524	0.000e+00	524	1.399e-01
525	0.000e+00	525	1.383e-01
526	0.000e+00	526	1.361e-01
527	0.000e+00	527	1.471e-01
528	0.000e+00	528	1.409e-01
529	0.000e+00	529	1.518e-01
530	0.000e+00	530	1.628e-01
531	0.000e+00	531	1.627e-01
532	0.000e+00	532	1.590e-01
533	0.000e+00	533	1.593e+00
534	0.000e+00	534	1.627e+00
535	0.000e+00	535	4.129e-01
536	0.000e+00	536	4.615e-01
537	0.000e+00	537	5.212e-01
538	0.000e+00	538	5.737e-01
539	0.000e+00	539	4.654e-01
540	0.000e+00	540	4.238e-01
541	0.000e+00	541	2.233e+00
542	0.000e+00	542	1.114e+00
543	0.000e+00	543	1.133e+00
544	0.000e+00	544	9.519e-01
545	0.000e+00	545	8.825e-01
546	0.000e+00	546	1.261e+00
547	0.000e+00	547	1.153e+00
548	0.000e+00	548	1.171e+00
549	0.000e+00	549	5.021e+00
550	0.000e+00	550	5.418e+00
551	0.000e+00	551	5.377e+00

Continua na próxima página

Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
552	0.000e+00	552	4.408e+00
553	0.000e+00	553	2.646e+00
554	0.000e+00	554	2.692e+00
555	0.000e+00	555	3.802e+00
556	0.000e+00	556	3.827e+00
557	0.000e+00	557	4.004e+00
558	0.000e+00	558	4.045e+00
559	0.000e+00	559	3.813e+00
560	0.000e+00	560	4.129e+00
561	0.000e+00	561	4.200e+00
562	0.000e+00	562	4.999e+00
563	0.000e+00	563	4.539e+00
564	0.000e+00	564	4.091e+00
565	0.000e+00	565	1.085e+01
566	0.000e+00	566	1.583e+01
567	0.000e+00	567	1.248e+01
568	0.000e+00	568	1.269e+01
569	0.000e+00	569	1.470e+01
570	0.000e+00	570	1.628e+01
571	0.000e+00	571	1.419e+01
572	0.000e+00	572	1.450e+01
573	0.000e+00	573	1.251e+02
574	0.000e+00	574	1.212e+02
575	0.000e+00	575	1.723e+02
576	0.000e+00	576	8.611e+01
577	0.000e+00	577	8.941e+01
578	0.000e+00	578	1.575e+02
579	0.000e+00	579	1.691e+02
580	0.000e+00	580	1.466e+02
581	1.171e+01	581	0.000e+00
582	0.000e+00	582	1.321e-01
583	0.000e+00	583	2.743e+00
584	0.000e+00	584	2.584e+00
585	0.000e+00	585	2.170e+00
586	0.000e+00	586	1.940e+00
587	0.000e+00	587	2.310e+00
588	0.000e+00	588	2.875e+00
589	3.164e+00	589	0.000e+00
590	2.542e+01	590	0.000e+00
591	1.909e+01	591	0.000e+00
592	0.000e+00	592	1.573e-01
593	0.000e+00	593	6.064e-01
594	0.000e+00	594	1.076e+00
595	0.000e+00	595	3.637e-01
596	1.073e+01	596	0.000e+00
597	0.000e+00	597	9.592e-02
598	0.000e+00	598	9.654e-02
599	0.000e+00	599	9.595e-02
600	0.000e+00	600	9.873e-02

Continua na próxima página

Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
601	0.000e+00	601	1.002e-01
602	0.000e+00	602	1.046e-01
603	0.000e+00	603	9.258e-02
604	0.000e+00	604	9.136e-02
605	0.000e+00	605	3.409e-01
606	0.000e+00	606	3.059e-01
607	0.000e+00	607	3.516e-01
608	0.000e+00	608	3.378e-01
609	0.000e+00	609	2.369e-01
610	0.000e+00	610	3.428e-01
611	0.000e+00	611	3.790e-01
612	0.000e+00	612	3.785e-01
613	0.000e+00	613	3.467e-01
614	0.000e+00	614	3.804e-01
615	0.000e+00	615	3.508e-01
616	0.000e+00	616	3.881e-01
617	0.000e+00	617	2.635e-01
618	0.000e+00	618	2.358e-01
619	0.000e+00	619	2.969e-01
620	0.000e+00	620	3.242e-01
621	0.000e+00	621	3.732e-01
622	0.000e+00	622	3.850e-01
623	0.000e+00	623	4.104e-01
624	0.000e+00	624	3.833e-01
625	0.000e+00	625	3.870e-01
626	0.000e+00	626	4.290e-01
627	0.000e+00	627	4.236e-01
628	0.000e+00	628	4.081e-01
629	0.000e+00	629	1.311e+00
630	0.000e+00	630	1.493e+00
631	0.000e+00	631	1.507e+00
632	0.000e+00	632	1.081e+00
633	0.000e+00	633	1.017e+00
634	0.000e+00	634	1.374e+00
635	0.000e+00	635	1.341e+00
636	0.000e+00	636	1.438e+00
637	0.000e+00	637	2.690e+00
638	0.000e+00	638	3.891e+00
639	0.000e+00	639	3.351e+00
640	0.000e+00	640	2.496e+00
641	0.000e+00	641	3.761e+00
642	0.000e+00	642	5.835e+00
643	0.000e+00	643	4.528e+00
644	0.000e+00	644	4.692e+00
645	0.000e+00	645	1.634e+00
646	0.000e+00	646	1.503e+00
647	0.000e+00	647	1.590e+00
648	0.000e+00	648	1.508e+00
649	0.000e+00	649	1.138e+00

Continua na próxima página

Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
650	0.000e+00	650	1.287e+00
651	0.000e+00	651	1.405e+00
652	0.000e+00	652	1.666e+00
653	4.222e+01	653	0.000e+00
654	9.990e+01	654	0.000e+00
655	0.000e+00	655	4.153e-03
656	0.000e+00	656	6.000e-02
657	3.106e+01	657	0.000e+00
658	0.000e+00	658	3.272e-01
659	0.000e+00	659	6.784e-03
660	1.967e+01	660	0.000e+00
661	1.880e+01	661	-2.427e-02
662	8.321e+01	662	-4.505e+04
663	1.257e+02	663	-4.505e+04
664	1.917e+01	664	-4.505e+04
665	4.347e+01	665	-3.843e-02
666	4.067e+00	666	-2.301e-01
667	3.807e+01	667	-6.847e-01
668	5.998e+01	668	-6.847e-01
669	2.979e-01	669	-6.847e-01
670	5.238e-01	670	1.639e-01
671	1.543e+01	671	-2.427e-02
672	2.419e+01	672	-4.505e+04
673	4.396e+01	673	-4.505e+04
674	2.963e+00	674	-4.505e+04
675	2.933e+01	675	-3.843e-02
676	5.576e+01	676	-3.068e-02
677	6.850e+01	677	1.890e-01
678	5.420e+01	678	-3.068e-02
679	7.950e+00	679	3.897e+01
680	2.200e+01	680	-3.068e-02
681	4.677e+01	681	-3.186e-01
682	4.529e+01	682	-3.186e-01
683	3.033e+01	683	-3.186e-01
684	3.664e+00	684	3.846e+01
685	1.018e+01	685	-7.703e-01
686	2.614e+01	686	-1.893e-02
687	6.458e+00	687	-4.713e-01
688	7.927e+00	688	-1.893e-02
689	6.380e-01	689	-8.972e+00
690	1.293e+01	690	-2.206e-01
691	5.701e+00	691	-4.757e-01
692	5.443e+00	692	-2.646e+00
693	7.506e+00	693	-4.757e-01
694	3.650e-01	694	-4.757e-01
695	5.783e+00	695	-4.955e-01
696	1.700e-01	696	-9.337e-01
697	1.974e+00	697	-5.669e-01
698	1.799e+00	698	-9.337e-01

Continua na próxima página

Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
699	3.170e-01	699	-5.669e-01
700	1.738e+00	700	-9.337e-01
701	7.963e-02	701	-6.058e-01
702	4.862e-01	702	-6.058e-01
703	1.768e+00	703	-6.058e-01
704	5.588e-04	704	-2.455e+02
705	4.593e-01	705	-2.500e+01
706	0.000e+00	706	-2.093e-01
707	1.009e+00	707	-1.222e+00
708	2.811e+00	708	-1.450e+00
709	1.224e-01	709	-8.419e+01
710	4.410e-02	710	-2.268e+00
711	2.512e+01	711	-5.797e-01
712	2.747e+01	712	-4.504e+00
713	2.126e+01	713	-2.713e-01
714	6.367e+00	714	-3.178e+01
715	9.452e+00	715	-9.077e+00
716	1.029e+02	716	-7.281e-04
717	1.022e+02	717	-6.741e-04
718	1.030e+02	718	-6.741e-04
719	1.022e+02	719	-6.523e-04
720	1.021e+02	720	-8.132e-04
721	9.988e+01	721	-8.044e-04
722	1.010e+02	722	-7.281e-04
723	1.015e+02	723	-6.523e-04
724	2.189e+01	724	-6.904e-03
725	2.171e+01	725	-7.135e-03
726	2.187e+01	726	-7.249e-03
727	2.171e+01	727	-7.016e-03
728	2.125e+01	728	-2.151e-02
729	2.165e+01	729	-1.440e-02
730	2.181e+01	730	-2.752e-03
731	2.187e+01	731	-2.782e-03
732	8.611e+02	732	-1.352e+03
733	8.524e+02	733	-1.397e+03
734	8.515e+02	734	-1.397e+03
735	8.499e+02	735	-1.352e+03
736	8.420e+02	736	-1.404e+03
737	8.538e+02	737	-1.389e+03
738	8.485e+02	738	-1.382e+03
739	8.438e+02	739	-1.382e+03
740	2.346e+02	740	-2.054e-02
741	2.301e+02	741	-2.123e-02
742	2.221e+02	742	-2.123e-02
743	2.327e+02	743	-2.054e-02
744	2.317e+02	744	-3.733e-02
745	2.381e+02	745	-3.692e-02
746	2.312e+02	746	-8.634e-03
747	2.294e+02	747	-8.729e-03

Continua na próxima página

Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
748	5.749e+03	748	-1.352e+03
749	5.705e+03	749	-1.397e+03
750	5.702e+03	750	-1.397e+03
751	5.701e+03	751	-1.352e+03
752	5.662e+03	752	-1.404e+03
753	5.666e+03	753	-1.389e+03
754	5.660e+03	754	-1.382e+03
755	5.653e+03	755	-1.382e+03
756	1.317e+03	756	-2.054e-02
757	1.308e+03	757	-2.123e-02
758	1.317e+03	758	-2.123e-02
759	1.307e+03	759	-2.054e-02
760	1.319e+03	760	-5.027e-02
761	1.307e+03	761	-4.973e-02
762	1.315e+03	762	-1.174e-02
763	1.317e+03	763	-8.729e-03
764	6.722e+02	764	-1.352e+03
765	6.698e+02	765	-1.397e+03
766	6.715e+02	766	-1.397e+03
767	6.740e+02	767	-1.352e+03
768	6.728e+02	768	-1.404e+03
769	6.540e+02	769	-1.389e+03
770	6.601e+02	770	-1.382e+03
771	6.659e+02	771	-1.382e+03
772	4.280e+01	772	-2.054e-02
773	4.243e+01	773	-2.123e-02
774	4.276e+01	774	-3.175e-02
775	4.243e+01	775	-3.072e-02
776	4.283e+01	776	-3.175e-02
777	4.242e+01	777	-3.175e-02
778	4.264e+01	778	-3.175e-02
779	4.275e+01	779	-3.175e-02
780	1.050e+03	780	-1.153e-03
781	1.043e+03	781	-9.822e-04
782	1.045e+03	782	-9.822e-04
783	1.047e+03	783	-9.505e-04
784	1.043e+03	784	-1.166e-03
785	1.026e+03	785	-1.153e-03
786	1.031e+03	786	-1.153e-03
787	1.036e+03	787	-9.505e-04
788	2.946e+01	788	4.918e-03
789	2.921e+01	789	5.082e-03
790	2.856e+01	790	5.082e-03
791	2.823e+01	791	4.918e-03
792	2.820e+01	792	2.456e-03
793	3.051e+01	793	-1.617e-04
794	2.968e+01	794	-1.617e-04
795	2.881e+01	795	-1.635e-04
796	9.478e+01	796	0.000e+00

Continua na próxima página

Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
797	1.303e+02	797	0.000e+00
798	1.658e+02	798	0.000e+00
799	1.790e+02	799	0.000e+00
800	2.145e+02	800	0.000e+00
801	2.500e+02	801	0.000e+00
802	2.509e+02	802	0.000e+00
803	2.160e+02	803	0.000e+00
804	2.600e+02	804	0.000e+00
805	3.421e+02	805	0.000e+00
806	4.241e+02	806	0.000e+00
807	4.546e+02	807	0.000e+00
808	5.366e+02	808	0.000e+00
809	6.187e+02	809	0.000e+00
810	6.208e+02	810	0.000e+00
811	5.401e+02	811	0.000e+00
812	4.567e+02	812	0.000e+00
813	5.811e+02	813	0.000e+00
814	7.055e+02	814	0.000e+00
815	7.518e+02	815	0.000e+00
816	8.762e+02	816	0.000e+00
817	1.001e+03	817	0.000e+00
818	1.004e+03	818	0.000e+00
819	8.815e+02	819	0.000e+00
820	3.499e+01	820	0.000e+00
821	4.603e+01	821	0.000e+00
822	5.707e+01	822	0.000e+00
823	6.117e+01	823	0.000e+00
824	7.221e+01	824	0.000e+00
825	8.325e+01	825	0.000e+00
826	8.354e+01	826	0.000e+00
827	7.269e+01	827	0.000e+00
828	1.269e+02	828	0.000e+00
829	1.744e+02	829	0.000e+00
830	2.219e+02	830	0.000e+00
831	2.396e+02	831	0.000e+00
832	2.871e+02	832	0.000e+00
833	3.346e+02	833	0.000e+00
834	3.358e+02	834	0.000e+00
835	2.891e+02	835	0.000e+00
836	1.533e+01	836	2.527e-03
837	2.120e+01	837	0.000e+00
838	2.747e+01	838	0.000e+00
839	1.510e+01	839	1.290e-02
840	3.733e+01	840	0.000e+00
841	1.382e+01	841	8.380e-04
842	1.533e+01	842	9.109e-04
843	1.510e+01	843	7.090e-03
844	7.522e+01	844	0.000e+00
845	7.100e+01	845	0.000e+00

Continua na próxima página

Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
846	6.619e+01	846	0.000e+00
847	6.759e+01	847	2.979e+04
848	6.541e+01	848	-1.819e-12
849	6.698e+01	849	8.135e+01
850	6.719e+01	850	1.651e+02
851	6.759e+01	851	1.502e+04
852	1.181e+02	852	0.000e+00
853	1.102e+02	853	0.000e+00
854	1.115e+02	854	0.000e+00
855	1.043e+02	855	2.979e+04
856	1.110e+02	856	0.000e+00
857	1.044e+02	857	8.135e+01
858	1.029e+02	858	1.651e+02
859	1.043e+02	859	1.502e+04
860	1.645e+01	860	0.000e+00
861	1.574e+01	861	0.000e+00
862	1.279e+01	862	0.000e+00
863	1.292e+01	863	2.979e+04
864	1.417e+01	864	0.000e+00
865	1.389e+01	865	8.135e+01
866	1.479e+01	866	1.651e+02
867	1.292e+01	867	1.502e+04
868	3.683e+01	868	6.751e-03
869	3.851e+01	869	0.000e+00
870	4.386e+01	870	0.000e+00
871	3.714e+01	871	1.901e-02
872	4.619e+01	872	0.000e+00
873	6.018e+01	873	0.000e+00
874	3.683e+01	874	2.339e-03
875	3.714e+01	875	1.033e-02
876	4.067e+00	876	0.000e+00
877	4.971e+00	877	-3.702e-03
878	4.971e+00	878	0.000e+00
879	4.971e+00	879	0.000e+00
880	4.067e+00	880	7.559e-02
881	4.067e+00	881	1.280e-01
882	4.067e+00	882	0.000e+00
883	4.067e+00	883	3.024e-02
884	3.807e+01	884	0.000e+00
885	3.866e+01	885	0.000e+00
886	4.514e+01	886	0.000e+00
887	3.807e+01	887	2.790e-01
888	4.378e+01	888	0.000e+00
889	3.807e+01	889	3.109e-01
890	4.653e+01	890	0.000e+00
891	3.807e+01	891	9.488e-02
892	5.998e+01	892	0.000e+00
893	5.998e+01	893	0.000e+00
894	5.998e+01	894	0.000e+00

Continua na próxima página

Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
895	5.998e+01	895	1.383e-01
896	6.306e+01	896	0.000e+00
897	5.998e+01	897	4.174e-01
898	5.998e+01	898	3.413e-02
899	5.998e+01	899	9.488e-02
900	2.979e-01	900	0.000e+00
901	3.641e-01	901	-3.393e-01
902	2.979e-01	902	0.000e+00
903	2.979e-01	903	6.790e-01
904	3.641e-01	904	-3.792e-03
905	2.979e-01	905	0.000e+00
906	2.979e-01	906	3.792e-03
907	2.979e-01	907	3.451e-01
908	5.238e-01	908	0.000e+00
909	6.402e-01	909	0.000e+00
910	5.238e-01	910	0.000e+00
911	6.402e-01	911	-1.372e-01
912	6.402e-01	912	-2.847e-02
913	6.402e-01	913	0.000e+00
914	6.402e-01	914	0.000e+00
915	5.238e-01	915	1.777e-03
916	1.565e+01	916	2.527e-03
917	1.460e+01	917	0.000e+00
918	1.427e+01	918	0.000e+00
919	1.562e+01	919	1.290e-02
920	1.530e+01	920	0.000e+00
921	1.425e+01	921	8.380e-04
922	1.554e+01	922	9.109e-04
923	1.482e+01	923	7.090e-03
924	2.237e+01	924	0.000e+00
925	2.138e+01	925	0.000e+00
926	2.097e+01	926	0.000e+00
927	2.101e+01	927	2.979e+04
928	2.092e+01	928	-1.819e-12
929	1.910e+01	929	8.135e+01
930	2.226e+01	930	1.651e+02
931	2.112e+01	931	1.502e+04
932	4.060e+01	932	0.000e+00
933	3.920e+01	933	0.000e+00
934	3.740e+01	934	0.000e+00
935	3.752e+01	935	2.979e+04
936	3.713e+01	936	-1.819e-12
937	3.691e+01	937	8.135e+01
938	3.983e+01	938	1.651e+02
939	3.805e+01	939	1.502e+04
940	3.058e+00	940	0.000e+00
941	2.892e+00	941	0.000e+00
942	2.812e+00	942	0.000e+00
943	2.412e+00	943	2.979e+04

Continua na próxima página

Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
944	2.678e+00	944	-1.819e-12
945	2.891e+00	945	8.135e+01
946	2.923e+00	946	1.651e+02
947	2.709e+00	947	1.502e+04
948	2.715e+01	948	6.751e-03
949	2.787e+01	949	0.000e+00
950	2.645e+01	950	0.000e+00
951	2.740e+01	951	1.901e-02
952	2.503e+01	952	0.000e+00
953	2.325e+01	953	0.000e+00
954	2.685e+01	954	2.339e-03
955	2.724e+01	955	1.033e-02
956	5.865e+01	956	0.000e+00
957	5.837e+01	957	0.000e+00
958	5.690e+01	958	0.000e+00
959	6.247e+01	959	0.000e+00
960	6.053e+01	960	-2.619e-02
961	6.261e+01	961	0.000e+00
962	6.495e+01	962	0.000e+00
963	5.281e+01	963	5.687e-02
964	6.751e+01	964	0.000e+00
965	6.457e+01	965	-2.897e-02
966	6.038e+01	966	0.000e+00
967	6.457e+01	967	-1.907e-01
968	6.540e+01	968	-2.619e-02
969	8.025e+01	969	0.000e+00
970	6.038e+01	970	0.000e+00
971	5.951e+01	971	5.687e-02
972	5.826e+01	972	0.000e+00
973	6.164e+01	973	0.000e+00
974	6.148e+01	974	0.000e+00
975	6.175e+01	975	0.000e+00
976	6.313e+01	976	-2.619e-02
977	6.167e+01	977	0.000e+00
978	5.703e+01	978	0.000e+00
979	5.591e+01	979	5.687e-02
980	7.478e+00	980	0.000e+00
981	6.432e+00	981	-3.897e+01
982	6.218e+00	982	0.000e+00
983	6.371e+00	983	0.000e+00
984	7.880e+00	984	-5.593e-03
985	9.373e+00	985	0.000e+00
986	8.042e+00	986	0.000e+00
987	5.286e+00	987	0.000e+00
988	2.223e+01	988	0.000e+00
989	2.223e+01	989	0.000e+00
990	2.307e+01	990	0.000e+00
991	2.425e+01	991	0.000e+00
992	2.589e+01	992	-2.619e-02

Continua na próxima página

Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
993	2.868e+01	993	0.000e+00
994	2.500e+01	994	0.000e+00
995	2.253e+01	995	5.687e-02
996	5.625e+01	996	0.000e+00
997	6.778e+01	997	0.000e+00
998	8.277e+01	998	0.000e+00
999	9.091e+01	999	0.000e+00
1000	6.679e+01	1000	2.719e-01

3 Variáveis Duais (Multiplicadores de Lagrange)

Tabela 2: Variáveis duais (y) do problema nesm

Coordenada y	Valor y (Dual)
1	-6.123e-03
2	-6.358e-03
3	-6.474e-03
4	-6.266e-03
5	-2.033e-02
6	-7.192e-03
7	-4.664e-03
8	-4.383e-03
9	1.352e+03
10	1.397e+03
11	1.397e+03
12	1.352e+03
13	1.404e+03
14	1.389e+03
15	1.382e+03
16	1.382e+03
17	1.352e+03
18	1.397e+03
19	1.397e+03
20	1.352e+03
21	1.404e+03
22	1.389e+03
23	1.382e+03
24	1.382e+03
25	1.352e+03
26	1.397e+03
27	1.397e+03
28	1.351e+03
29	1.404e+03
30	1.389e+03
31	1.381e+03
32	1.381e+03

Continua na próxima página

Coordenada y	Valor y (Dual)
33	-5.965e-02
34	-5.843e-02
35	-5.843e-02
36	-5.655e-02
37	-6.925e-02
38	-1.174e-02
39	-5.633e-02
40	-5.550e-02
41	0.000e+00
42	0.000e+00
43	0.000e+00
44	0.000e+00
45	0.000e+00
46	0.000e+00
47	0.000e+00
48	0.000e+00
49	0.000e+00
50	0.000e+00
51	0.000e+00
52	0.000e+00
53	0.000e+00
54	0.000e+00
55	0.000e+00
56	0.000e+00
57	-2.182e+03
58	-2.157e+03
59	-2.157e+03
60	-2.182e+03
61	-2.153e+03
62	-2.161e+03
63	-2.166e+03
64	-2.166e+03
65	-2.376e+03
66	-2.376e+03
67	-2.376e+03
68	-2.376e+03
69	-2.376e+03
70	-2.376e+03
71	-2.376e+03
72	-2.376e+03
73	0.000e+00
74	0.000e+00
75	0.000e+00
76	0.000e+00
77	0.000e+00
78	0.000e+00
79	0.000e+00
80	0.000e+00
81	-1.775e+03

Continua na próxima página

Coordenada y	Valor y (Dual)
82	-1.775e+03
83	-1.775e+03
84	-1.775e+03
85	-1.775e+03
86	-1.775e+03
87	-1.775e+03
88	-1.775e+03
89	-3.327e+02
90	-1.479e+02
91	-1.479e+02
92	-3.327e+02
93	-1.164e+02
94	-1.790e+02
95	-2.093e+02
96	-2.093e+02
97	0.000e+00
98	0.000e+00
99	0.000e+00
100	0.000e+00
101	0.000e+00
102	0.000e+00
103	0.000e+00
104	0.000e+00
105	0.000e+00
106	0.000e+00
107	0.000e+00
108	0.000e+00
109	0.000e+00
110	0.000e+00
111	0.000e+00
112	0.000e+00
113	0.000e+00
114	0.000e+00
115	0.000e+00
116	0.000e+00
117	0.000e+00
118	0.000e+00
119	0.000e+00
120	0.000e+00
121	0.000e+00
122	0.000e+00
123	0.000e+00
124	0.000e+00
125	0.000e+00
126	0.000e+00
127	0.000e+00
128	-3.265e+02
129	-1.631e+02
130	0.000e+00
Continua na próxima página	

Coordenada y	Valor y (Dual)
131	-3.269e+02
132	-8.058e+01
133	0.000e+00
134	0.000e+00
135	7.281e-04
136	6.741e-04
137	6.741e-04
138	6.523e-04
139	8.132e-04
140	8.044e-04
141	7.281e-04
142	6.523e-04
143	1.352e+03
144	1.397e+03
145	1.397e+03
146	1.352e+03
147	1.404e+03
148	1.389e+03
149	1.382e+03
150	1.382e+03
151	1.352e+03
152	1.397e+03
153	1.397e+03
154	1.352e+03
155	1.404e+03
156	1.389e+03
157	1.382e+03
158	1.382e+03
159	1.352e+03
160	1.397e+03
161	1.397e+03
162	1.352e+03
163	1.404e+03
164	1.389e+03
165	1.382e+03
166	1.382e+03
167	1.153e-03
168	9.822e-04
169	9.822e-04
170	9.505e-04
171	1.166e-03
172	1.153e-03
173	1.153e-03
174	9.505e-04
175	6.904e-03
176	7.135e-03
177	7.249e-03
178	7.016e-03
179	2.151e-02
Continua na próxima página	

Coordenada y	Valor y (Dual)
180	1.440e-02
181	2.752e-03
182	2.782e-03
183	2.054e-02
184	2.123e-02
185	2.123e-02
186	2.054e-02
187	3.733e-02
188	3.692e-02
189	8.634e-03
190	8.729e-03
191	2.054e-02
192	2.123e-02
193	2.123e-02
194	2.054e-02
195	5.027e-02
196	4.973e-02
197	1.174e-02
198	8.729e-03
199	2.054e-02
200	2.123e-02
201	3.175e-02
202	3.072e-02
203	3.175e-02
204	3.175e-02
205	3.175e-02
206	3.175e-02
207	-4.918e-03
208	-5.082e-03
209	-5.082e-03
210	-4.918e-03
211	-2.456e-03
212	1.617e-04
213	1.617e-04
214	1.635e-04
215	7.281e-04
216	6.741e-04
217	6.741e-04
218	6.523e-04
219	8.132e-04
220	8.044e-04
221	7.281e-04
222	6.523e-04
223	1.352e+03
224	1.397e+03
225	1.397e+03
226	1.352e+03
227	1.404e+03
228	1.389e+03

Continúa na próxima página

Coordenada y	Valor y (Dual)
229	1.382e+03
230	1.382e+03
231	1.352e+03
232	1.397e+03
233	1.397e+03
234	1.352e+03
235	1.404e+03
236	1.389e+03
237	1.382e+03
238	1.382e+03
239	1.352e+03
240	1.397e+03
241	1.397e+03
242	1.352e+03
243	1.404e+03
244	1.389e+03
245	1.382e+03
246	1.382e+03
247	1.153e-03
248	9.822e-04
249	9.822e-04
250	9.505e-04
251	1.166e-03
252	1.153e-03
253	1.153e-03
254	9.505e-04
255	9.204e-04
256	9.511e-04
257	9.511e-04
258	9.204e-04
259	2.823e-03
260	5.175e-03
261	5.175e-03
262	5.232e-03
263	-5.671e-03
264	-5.860e-03
265	-4.962e-03
266	-4.802e-03
267	2.823e-03
268	5.175e-03
269	5.175e-03
270	5.232e-03
271	9.204e-04
272	9.511e-04
273	9.511e-04
274	9.204e-04
275	2.823e-03
276	5.175e-03
277	5.175e-03

Continua na próxima página

Coordenada y	Valor y (Dual)
278	5.232e-03
279	-1.169e+00
280	-1.208e+00
281	-1.734e-04
282	-1.678e-04
283	-5.145e-04
284	0.000e+00
285	0.000e+00
286	0.000e+00
287	9.204e-04
288	9.511e-04
289	9.511e-04
290	9.204e-04
291	2.823e-03
292	5.175e-03
293	5.175e-03
294	5.232e-03
295	9.559e-03
296	9.878e-03
297	9.878e-03
298	9.559e-03
299	2.931e-02
300	4.252e-03
301	4.252e-03
302	3.635e-03
303	9.559e-03
304	9.878e-03
305	9.878e-03
306	9.559e-03
307	2.931e-02
308	4.252e-03
309	4.252e-03
310	3.635e-03
311	9.559e-03
312	9.878e-03
313	9.878e-03
314	9.559e-03
315	2.931e-02
316	4.252e-03
317	4.252e-03
318	3.635e-03
319	-1.154e+00
320	2.388e-02
321	2.388e-02
322	2.311e-02
323	7.087e-02
324	4.252e-03
325	4.252e-03
326	4.298e-03
Continua na próxima página	

Coordenada y	Valor y (Dual)
327	2.311e-02
328	2.388e-02
329	2.388e-02
330	2.311e-02
331	7.087e-02
332	4.252e-03
333	4.252e-03
334	4.298e-03
335	5.679e-04
336	5.599e-04
337	5.599e-04
338	5.599e-04
339	4.932e-04
340	4.511e-04
341	5.679e-04
342	5.599e-04
343	1.414e-02
344	1.461e-02
345	1.461e-02
346	1.414e-02
347	4.336e-02
348	4.288e-02
349	3.140e-02
350	3.175e-02
351	5.679e-04
352	5.786e-04
353	5.786e-04
354	5.599e-04
355	1.253e-02
356	1.758e-02
357	1.539e-02
358	3.436e-03
359	2.691e-01
360	2.781e-01
361	2.781e-01
362	2.691e-01
363	8.254e-01
364	8.164e-01
365	8.164e-01
366	8.253e-01
367	6.619e-03
368	6.839e-03
369	6.839e-03
370	6.619e-03
371	2.030e-02
372	2.008e-02
373	2.008e-02
374	8.140e-03
375	1.427e-02

Continua na próxima página

Coordenada y	Valor y (Dual)
376	1.475e-02
377	1.475e-02
378	1.427e-02
379	4.377e-02
380	4.329e-02
381	2.323e-02
382	1.070e-02
383	7.937e-02
384	8.201e-02
385	8.201e-02
386	7.937e-02
387	7.584e-02
388	4.329e-02
389	4.329e-02
390	4.377e-02
391	1.427e-02
392	1.475e-02
393	1.475e-02
394	1.427e-02
395	4.377e-02
396	4.329e-02
397	2.323e-02
398	1.070e-02
399	1.427e-02
400	1.475e-02
401	1.475e-02
402	1.427e-02
403	2.348e-02
404	2.323e-02
405	2.323e-02
406	2.348e-02
407	1.487e-02
408	1.536e-02
409	1.536e-02
410	1.487e-02
411	1.070e-02
412	1.058e-02
413	1.058e-02
414	1.070e-02
415	2.801e-02
416	1.757e-02
417	1.757e-02
418	1.701e-02
419	5.215e-02
420	3.175e-02
421	2.801e-02
422	1.701e-02
423	1.701e-02
424	1.757e-02
Continua na próxima página	

Coordenada y	Valor y (Dual)
425	1.757e-02
426	1.701e-02
427	5.215e-02
428	3.334e-02
429	3.334e-02
430	3.371e-02
431	2.801e-02
432	1.757e-02
433	1.757e-02
434	1.701e-02
435	5.215e-02
436	3.175e-02
437	2.801e-02
438	1.701e-02
439	1.701e-02
440	1.757e-02
441	1.757e-02
442	1.701e-02
443	5.215e-02
444	3.334e-02
445	3.334e-02
446	3.371e-02
447	2.801e-02
448	1.757e-02
449	1.757e-02
450	1.701e-02
451	2.801e-02
452	1.217e-02
453	1.217e-02
454	1.230e-02
455	1.818e-02
456	1.878e-02
457	1.878e-02
458	1.817e-02
459	5.573e-02
460	0.000e+00
461	0.000e+00
462	0.000e+00
463	1.817e-02
464	1.878e-02
465	1.878e-02
466	1.817e-02
467	5.573e-02
468	0.000e+00
469	0.000e+00
470	0.000e+00
471	1.818e-02
472	1.878e-02
473	1.878e-02

Continua na próxima página

Coordenada y	Valor y (Dual)
474	1.817e-02
475	5.573e-02
476	0.000e+00
477	0.000e+00
478	0.000e+00
479	7.366e+00
480	7.611e+00
481	7.611e+00
482	7.366e+00
483	1.019e+01
484	5.997e+00
485	5.997e+00
486	6.063e+00
487	7.501e-01
488	6.989e-01
489	6.989e-01
490	6.764e-01
491	4.929e-01
492	0.000e+00
493	0.000e+00
494	0.000e+00
495	6.279e-03
496	6.489e-03
497	6.489e-03
498	6.279e-03
499	1.926e-02
500	1.905e-02